
Part I. For fill-in-the-blank questions, each blank is worth a point. For true-or-false questions, write only T or F. For identification questions, each item is worth a point.

1. _____ of a rigid body represents it as being isolated from its surroundings, replacing physical supports with forces and couple moments.

The Free-Body Diagram (FBD)

2. True or False. An ideal truss does not have three-force members.

T

3. For 3D systems, the internal moment has two components: _____ and _____, whose axes are tangent and normal to the cross-section, respectively.

Bending and Torsion

4. A body is said to be _____ constrained if it experiences fewer reactive forces than applicable equations of static equilibrium.

Under

5. True or False. A beam supported by a pin (2), a rocker (1), and a roller (1) is statically determinate.

F

6. For a simply supported 13-member plane truss to be considered simple, it must have ____ joints.

8

7. True or False. A frictionless ball-and-socket resists both rotation and translation.

F

8. Body where all reactive forces intersect at one point or pass through a common axis is _____ Constrained.

Improperly

9. True or False. A stable body requires that the lines of action of the reactive forces do not intersect at a common axis and are not parallel to one another.

T

10. For internal loadings of 2D systems, the _____ force and the _____ force are tangent and normal to the cross-section, respectively.

Normal force and Shear force

Scenario 1

A 1.1-m long cantilever beam is subject to a 2-kN weight at its free end.

11. True or False. This scenario is statistically determinate.

T

12. The support reactions have magnitudes of ____ kN and ____ kN·m.

**2 kN (vertical reaction at the fixed support) and
2.2 kN·m (reaction moment at the fixed support)**

13. The maximum bending moment developed is ____ kN·m in magnitude and occurs ____ m from the free end..

**2.2 kN·m
1.1 m (at the fixed point)**

14. The maximum shear force developed is ____ kN-m in magnitude and occurs ____ m from the free end (the shear is constant along the cantilever).

**2 kN-m
0-1.1 m**

15. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree ____.

0

16. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree ____.

1

Scenario 2

A 1-m long beam carries a load uniformly distributed throughout its length (1 kN/m). Assume a roller support 0.1 m from the left end and a pin support at the right end.

17. The roller support develops a force of _____ kN.

0.556

18. The pin support develops a force of _____ kN.

0.444

19. The bending moment developed at the right end is _____ kN·m.

0

20. The shear force developed just to the left of the pin is _____ kN.

0.444

21. The shear force developed just to the right of the pin is _____ kN.

0

22. True or False. There is zero shear force developed at a point 0.9 m from the right end.

False

23. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree _____.

1

24. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree _____.

2

Part II:

Instructions: Where applicable, provide force diagrams and indicate the sense of member forces. Enclose final answers in boxes, with non-integer values rounded to three (3) decimal places.

Problem I



Problem 1: Truss Analysis

Given Parameters:

Loads: $p_1 = 5 \text{ kN}$, $p_2 = 6 \text{ kN}$, $p_3 = 7 \text{ kN}$.

Geometry:

Panel width $d = 2 \text{ ft}$, Truss height $h = 3 \text{ ft}$.

Slope Properties:

For the first sloped segment ($A \rightarrow G \rightarrow F$), the truss spans a horizontal distance of $2d = 4 \text{ ft}$ and rises by $d = 2 \text{ ft}$.

This forms a slope triangle with:

- Horizontal components = 2
- Vertical = 1
- Hypotenuse = $2^2 + 1^2 = 5$
- $\cos = 2/5 = 0.4$, $\sin = 1/5 = 0.2$

Crafting the Answer & Enchanting the Logic

Decomposition of Loads

The loads p_1 , p_2 , p_3 act perpendicular to the top chord, and since the chord is inclined, the loads are decomposed into horizontal (x) and vertical (y) components based on the truss's slope.

$$p_1 = 515 \text{ i} - 25 \text{ j} = 2.236 \text{ i} - 4.472 \text{ j kN}$$

$$p_2 = 615 \text{ i} - 25 \text{ j} = 2.683 \text{ i} - 5.367 \text{ j kN}$$

$$p_3 = 715 \text{ i} - 25 \text{ j} = 3.130 \text{ i} - 6.261 \text{ j kN}$$

Method of Sections (Part a)

To calculate the force in member FG, we introduce a vertical cut through the truss that passes through FG, CF, and CD. The analysis is then carried out on the left portion of the structure, containing joints A, B, C, and G. Moment about Joint C to find F_{FG} :

Taking moments about C eliminates unknowns F_{CG} and F_{CF} .

$$Mp_1 = p_1(4) = 4.472(4) = 17.888 \text{ kN-ft (CCW).}$$

Load p2 (at G): G is located at $x = 2$, $y = 1$ relative to A. Relative to C ($x = 4$, $y = 0$), G is at $x = -2$, $\Delta y = 1$.

$$Mp_2 = (p_2y)(\Delta x) - (p_2x)(\Delta y) = (-5.367)(-2) - (2.683)(1) = 10.734 - 2.683 = 8.051 \text{ kN-ft (CCW).}$$

Member FFG (at G): Force acts along the top chord. The perpendicular distance from C to the line AG is $d = 45 \approx 1.789$ ft. $M_{FG} = -1.789 F_{FG}$.

$$MC = 17.888 + 8.051 - 1.789 F_{FG} = 0$$
$$F_{FG} = 14.500 \text{ kN (T)}$$

B. Moment about Joint F to find F_{CD} :

Intersection of F_{FG} and F_{CF} is at F.

Moment of p_1 (at A) about F: $M_C = 22.360 \text{ kN-ft (CCW).}$

Moment of p_2 (at G) about F: $M_{p_2} = 13.417 \text{ kN-ft (CCW).}$

Moment of FCD: Force is horizontal. Lever arm is vertical height of $F = 2$ ft. $M_{MD} = -F_{CD}(2)$.

$$\sum M_F = 22.360 + 13.417 - 2 F_{CD} = 0$$
$$F_{CD} = 17.889 \text{ kN (C)}$$

C. Vertical Summation to find F_{CF} :

Summing forces on the left section:

$$\sum F_Y = -9.839 + 14.500(0.4472) + F_{CF} = 0$$
$$F_{CF} = 3.354 \text{ kN (T)}$$

Method of Joints (Part b)

Isolate Joint F. Applied Forces: Load p_2 ($x = 3.130$, $y = -6.261$).

Member Forces: F_{FG} (Known T), F_{CF} (Unknown), F_{DF} (Unknown)

$$\sum F_X = 3.130 - 12.97 + 0.894F_{EF} + 0.707F_{DF} = 0$$

$$\sum F_Y = -6.261 - 6.48 - 3.354 + 0.447F_{EF} - 0.707F_{DF} = 0$$

Solving simultaneous equations:

$$F_{EF} = 19.330 \text{ kN (T)}$$

$$F_{DF} = 10.520 \text{ kN (C)}$$

Stability Analysis (Part c)

Recommended Support Configuration:

- **Pin at D**
- **Roller at E** (positioned to resist horizontal movement)

Reasoning:

1. **Translational Stability:**

The inclined loads introduce strong horizontal force components.

To properly restrain the structure in both the x- and y-directions, a **pin at point D** is necessary since it can provide reactions in both axes (Dx and Dy).

2. **Rotational Stability:**

The applied cantilever loads generate a considerable overturning moment.

If a **vertical roller** were placed at E, its reaction would align with the vertical reaction at D, meaning it would offer **no moment resistance**, making the structure unstable.

To resist rotation, the support at E must supply a **horizontal reaction**, forming a force couple with the pin at D. This horizontal reaction pair counters the moment and keeps the structure stable.

Zero-Force Members (Part d)

Contingency: Joint A is unloaded ($P_1 = 0$).

• **Joint A:** Connects AB and AG (non-collinear). With no external load, equilibrium requires $F_{AB} = 0$ and $F_{AG} = 0$.

• **Joint B:** Connects AB, BC, and BG. Since $F_{AB} = 0$, horizontal equilibrium requires $F_{BC} = 0$.

Vertical equilibrium requires $F_{BG} = 0$.

Answer: The zero-force members are **AB**, **AG**, **BG**, and **BC**.

Support Reactions (Part e)

Reaction at E: Balances F_{EF} directly.

$$R_E = F_{EF} = 19.330 \text{ kN}$$

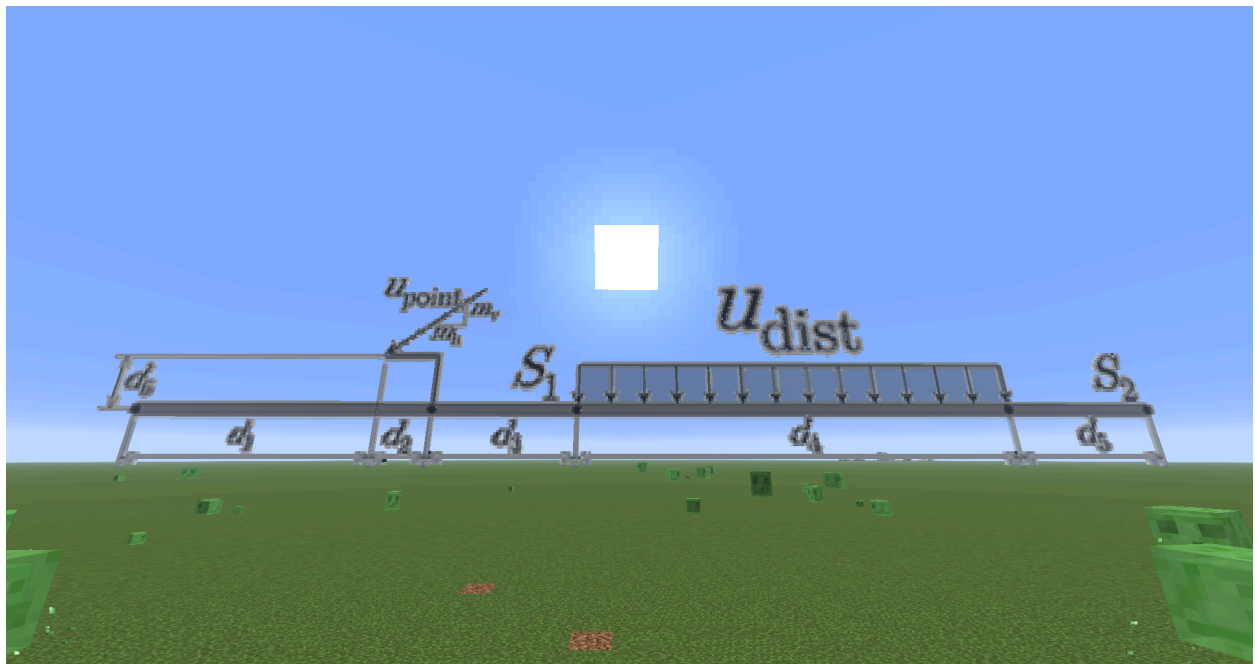
Reaction at D: Balances F_{CD} (Horizontal) and F_{DF} (Diagonal).

$$F_{DX} = F_{CD} + F_{DF} \cos(45^\circ) = 17.889 - 10.520(0.707) = 10.45 \text{ kN}$$

$$R_{Dy} = -F_{DF} \sin(45^\circ) = -10.520(0.707) = -7.44 \text{ kN}$$

$$|R_D| = \sqrt{10.45^2 + 7.442^2} = 12.830 \text{ kN}$$

Problem II



Problem 2: Beam Analysis

PARAMETERS (x represents distance from the left end of the beam):

- **Start of Beam:**
x = 0 ft.
- **Point (B):**
Located at $d_1 + d_2 = 3.2 + 0.8 = 4.0$ ft.
- **Pin (S1):**
Located at $x_B + d_3 = 4.0 + 2.0 = 6.0$ ft.
- **End of Distributed Load (D):**
Located at $x_{S1} + d_4 = 6.0 + 6.0 = 12.0$ ft.
- **Roller (S2):**
Located at $x_D + d_5 = 12.0 + 2.0 = 14.0$ ft.

A bracket at $x = 4.000$ applies the concentrated force of 5 kips.
The load slope is 3 Vertical : 4 Horizontal.

$$F_x = 5 \times \frac{4}{5} = 4 \text{ kips} \quad (\text{Pointing Left})$$

$$F_y = 5 \times \frac{3}{5} = 3 \text{ kips} \quad (\text{Pointing Down})$$

We relocate these forces, moving them from the bracket tip onto the beam axis where ($x=4.0$).

- horizontal distance:

$$d_2 = 0.8 \text{ ft.}$$

- vertical height:

$$d_6 = 0.6 \text{ ft.}$$

Counter-Clockwise (CCW, positive(+))

$$M_{\text{applied}} = (3 \text{ kips}) (0.8 \text{ ft}) + (4 \text{ kips}) (0.6 \text{ ft})$$

$$M_{\text{applied}} = 2.4 + 2.4 = 4.8 \text{ k-ft. (CCW)}$$

Finding external reactions at S1 and S2 and sum of moments about S1:

Moment from Point Load ($x=4$)

F: 3 kips

D: $6 - 4 = 2 \text{ ft.}$

$$+ (3)(2) = +6.$$

Concentrated Couple ($x=4$)

$$+4.8$$

Moment from Distributed Load

F: $2(6) = 12 \text{ kips.}$

D: 3 ft.

$$-(12)(3) = -36.$$

Moment from reaction

F: S2($x = 4$)

D: $(14 - 6) = 8 \text{ ft.}$

$$+S_2y(8)$$

$$\sum M_{S_1} = 6 + 4.8 - 36 + 8S_{2y} = 0$$

$$-25.2 + 8S_{2y} = 0 \rightarrow S_{2y} = \frac{25.2}{8} = 3.15 \text{ kips}$$

Sum of Vertical Forces:

$$\sum F_y = S_{1y} + S_{2y} - F_y - W_{dist} = 0$$

$$S_{1y} + 3.15 - 3 - 12 = 0$$

$$S_{1y} + 3.15 - 15 = 0 \rightarrow S_{1y} = 15 - 3.15 = 11.85 \text{ kips}$$

Sum of Horizontal Forces:

$$\sum F_x = S_{1x} - F_x = 0 \rightarrow S_{1x} - 4 = 0 \rightarrow S_{1x} = 4.00 \text{ kips}$$

4. Shear and Moment Functions (Piecewise Derivation)

Region 1: $0 \leq x < 4$ (Free End to Load)

- No external loads exist in this region.
- $V(x) = 0$
- $M(x) = 0$

Region 2: $4 < x < 6$ (Load to Support S_1)

- **Shear:** The vertical load of 3 kips pushes down at $x = 4$.

$$V(x) = 0 - 3 = -3$$

- **Moment: Integrate shear:** $\int -3 dx = -3x + C_1$.
- **Boundary Condition:** At $x = 4$, there is a concentrated moment of 4.8 (CCW). In beam conventions, a CCW external moment causes a drop (negative jump) in the diagram.

$$M(4^+) = M(4^-) - 4.8 = 0 - 4.8 = -4.8$$

$$\text{Substitute into the equation: } -3(4^-) + C_1 = -4.8 \rightarrow -12 + C_1 = -4.8 \rightarrow C_1 = 7.2.$$

$$M(x) = -4x + 7.2$$

Region 3: $6 < x < 12$ (Support S_1 to End of Dist. Load)

Shear: At $x = 6$, the reaction $S_{1y} = 11.85$ pushes up.

$$V(6^+) = V(6^-) + 11.85 = -3 + 11.85 = 8.85$$

The distributed load $w = 2$ k/ft reduces shear linearly from $x = 6$.

$$V(x) = 8.85 - 2(x - 6) = 8.85 - 2x + 12 = 20.85 - 2x$$

Moment: Integrate shear:

$$M(x) = \int (20.85 - 2x) dx = 20.85x - x^2 + C_2$$

Boundary Condition: Moment must be continuous at the pin (no applied couple at $x = 6$).

Check $M(6)$ from Region 2: $M(6) = -3(6) + 7.2 = -10.8$.

Substitute into Region 3 equation:

$$20.85(6) - (6)^2 + C_2 = -10.8$$

$$125.1 - 36 + C_2 = -10.8$$

$$89.1 + C_2 = -10.8 \Rightarrow C_2 = -99.9$$

$$M(x) = -x^2 + 20.85x - 99.9$$

Region 4: $12 < x \leq 14$ (End of Dist. Load to Roller)

Shear: The distributed load ends. Shear becomes constant. Check $V(12)$ from Region 3:

$$V(12) = 20.85 - 2(12) = -3.15$$

$$V(x) = -3.15$$

Moment: Integrate shear:

$$M(x) = \int -3.15 dx = -3.15x + C_3$$

Boundary Condition: Continuity at $x = 12$. Check $M(12)$ from Region 3:

$$M(12) = -(12)^2 + 20.85(12) - 99.9$$

$$M(12) = -144 + 250.2 - 99.9 = 6.3$$

Substitute into Region 4 equation:

$$-3.15(12) + C_3 = 6.3$$

$$-37.8 + C_3 = 6.3 \Rightarrow C_3 = 44.1$$

$$M(x) = -3.15x + 44.1$$

Shear Force $V(x)$ (kips):

$$V(x) = \begin{cases} 0 & 0 \leq x < 4 \\ -3 & 4 < x < 6 \\ 20.85 - 2x & 6 < x < 12 \\ -3.15 & 12 < x \leq 14 \end{cases}$$

Bending Moment $M(x)$ (k-ft):

$$M(x) = \begin{cases} 0 & 0 \leq x < 4 \\ -3x + 7.2 & 4 < x < 6 \\ -x^2 + 20.85x - 99.9 & 6 < x < 12 \\ -3.15x + 44.1 & 12 < x \leq 14 \end{cases}$$

ZERO SHEAR LOCATION: Occurs in Region 3 where $V(x)=0$.

$$20.85 - 2x = 0$$

$$2x = 20.85$$

$$x=10.425 \text{ ft}$$

MAXIMUM BENDING MOMENT: The maximum moment occurs where shear is zero ($x = 10.425$).

$$M_{\max} = -(10.425)^2 + 20.85(10.425) - 99.9$$

$$= -108.68 + 217.36 - 99.9$$

$$= 8.78 \text{ k-ft}$$

Closure Check at End ($x = 14$):

$$M(14) = -3.15(14) + 44.1$$

$$= -44.1 + 44.1$$

$$= 0$$